Ali Shafiee

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Codeforces | LinkedIn | IOI Profile | Google Scholar

IOI silver medalist and ICPC World Finalist. PhD researcher in probability, stochastic control, and algorithmic game theory.

Education

Institute of Science and Technology Austria (ISTA) — Vienna, Austria
PhD in Computer Science, advisor Prof. Krishnendu Chatterjee — *Sep. 2025 – present*

Sharif University of Technology — Tehran, Iran
BSc in Computer Engineering — *Sep. 2018 – Aug. 2023*

Competitive Programming

- **Codeforces**: **top 50 worldwide in 2019**; International Grandmaster, peak rating 2800 (handle `ToTLeS`)
- **Silver Medal, International Olympiad in Informatics (IOI)**, 2017: 76th of 300+ contestants from 80 countries
- **33rd Place, ACM-ICPC World Finals**, Moscow 2021: among 10,000+ teams from 110 countries
- **Gold Medal, ACM-ICPC West Asia Regional Contest**: 1st place 2019, 2nd 2018, 3rd 2020; 100+ teams each year
- **Silver Medal, Asia-Pacific Informatics Olympiad (APIO)**, 2018 and 2017: 20+ participating countries
- **Gold Medal, Iranian National Olympiad in Informatics**, 2016: top 8 of 10,000+ students nationally
- **Bronze Medal, Iranian Mathematical Olympiad**, 2017: top 80 of 20,000+ students nationally

Research

Complexity and algorithms for **sequential decision-making under uncertainty**: Markov decision processes with partial observation and with adversarial model uncertainty, plus **algorithmic game theory**. Author lists are alphabetical, per convention in the field.

- Randomise Alone, Reach as a Team. L. Brice, T. A. Henzinger, A. Montaseri, **A. Shafiee**, K. S. Thejaswini. *CAV 2026*.
- Qualitative Analysis of ω-Regular Objectives on Robust MDPs. A. Asadi, K. Chatterjee, E. Goharshady, M. Karrabi, **A. Shafiee**. *AAAI 2026*.
- Limit-Sure Reachability for Small-Memory Policies in POMDPs is NP-Complete. A. Asadi, K. Chatterjee, R. Saona, **A. Shafiee**. *UAI 2025*.
- *Under review:* On the Complexity of Discounted Robust MDPs with L_p Uncertainty Sets (NeurIPS 2026); Mechanism Design for Multi-Resource Nakamoto Consensus (WINE 2026); Quantitative Analysis of ω-Regular Robust MDPs (AAAI 2027).

Industry Experience

Quera, Iran's largest competitive-programming and technical-education platform — Tehran, Iran
Contest Manager → Content Manager → Director of CS Education → Business Line Manager — *Mar. 2019 – Aug. 2025*

- Owned a business line end to end (Oct. 2024 – Aug. 2025); previously led a 40-person division spanning six technical teams (back-end, front-end, data, blockchain, algorithms, fundamentals) and set the education product roadmap.
- Designed and shipped 12+ bootcamps and courses in C++, Python, SQL, Django, Linux/Git, data analysis, machine learning, and deep learning.
- Ran 20+ large-scale programming contests, including sponsored hiring hackathons with corporate partners.

Research Experience

Research Intern & Online Collaborator, ISTA — Vienna, Austria
Advisor: Prof. Krishnendu Chatterjee — *Apr. 2024 – Dec. 2024*

- Proved that limit-sure reachability for small-memory POMDP policies is NP-complete, settling the complexity of a long-open case (UAI 2025).
- Established complexity bounds and algorithms for robust MDPs with ω-regular and discounted objectives (three further papers); implemented and benchmarked the resulting solvers in C++/Python across randomly generated and structured MDP families.

Research Assistant, Sharif University of Technology — Tehran, Iran
Advisor: Prof. Mohammad Ali Abam — *Jun. 2023 – Aug. 2023*

- Round- and memory-efficient minimum cut and maximal matching algorithms in the Massive Parallel Computation (MPC) and Adaptive MPC models.

Teaching & Service

- **Scientific Committee**, ACM-ICPC West Asia Regional Contest (2024 – present); **Host Scientific Committee**, IOI 2019, Baku
- **Problem-setter** (2017 – present): Iranian National Olympiad in Informatics, Codecup, and Quera contests
- **Teaching Assistant**, Sharif University: Design of Algorithms, Data Structures & Algorithms, Fundamentals of Programming
- **Olympiad coach** (2017 – 2023): trained 50+ students for the Iranian National Olympiad in Informatics and the IOI

Skills

Programming: C++, Python, Rust · **Numerical & ML**: NumPy, SciPy, pandas, Matplotlib, scikit-learn, PyTorch, JAX
Tools: Git, Linux/Bash, SQL, Docker, Jupyter, LaTeX · **Mathematics**: probability, stochastic processes, Markov decision processes, optimization, game theory
Languages: Persian (native), English (TOEFL 110/120)